



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

## SAFETY DATA SHEET

### Company Identification

Planter United Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510; Fax: +66(0)-2721-3515

### 1. CHEMICAL IDENTIFICATION

#### Product Identification

Product name: abamectin 1.8% W/V EC

Product use: Insecticide

#### Relevant identified uses of the substance/mixture and uses advised against:

Used for the control of certain mites and insect pests on various crops.

### 2. HAZARD IDENTIFICATION

#### Poisoning Symptoms:

No case of human poisoning due to this product is on record. The poisoning symptoms observed in laboratory studies include decreased activity, slow rate of breathing, dilation of the pupils, musculotremors and lack of coordination.

### 3. COMPOSITION/INFORMATION ON INGREDIENT

Substances: Formulation

Chemical name: 5-O-demethylavermectin A1a (i) mixture with 5-O-demethyl-25-de(1-methylpropyl)-25-(1-methylethyl)avermectin A1a (ii)

Common name: abamectin

CAS Number: 71751-41-2

Ingredients and composition: Active ingredient - abamectin 1.8% W/V

Chemical family: Avermectin

Formula:  $C_{48}H_{72}O_{14}$  (avermectin B<sub>1a</sub>);  $C_{47}H_{70}O_{14}$  (avermectin B<sub>1b</sub>)

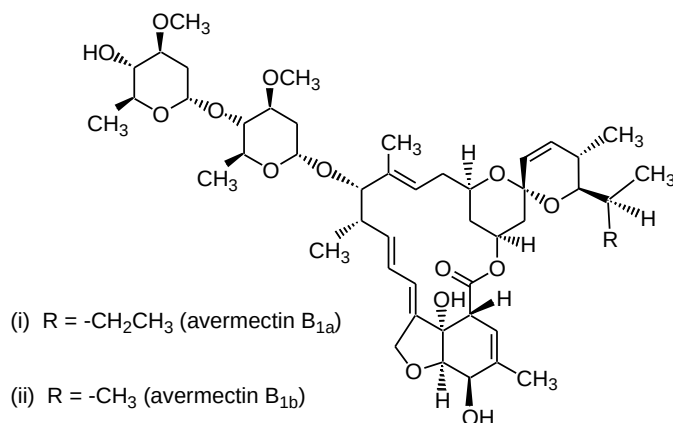


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#### 4. FIRST AID MEASURES

##### Description of first aid measures

In case of swallowed:

If swallowed contact a doctor or Poisons Information Centre.

In case of eyes contact:

If product gets in eyes hold eyes open, flush with running water for at least 15 minutes and see a doctor.

In case of skin contact:

If skin contact occurs, remove contaminated clothing and wash affected areas thoroughly with soap and water. Wash contaminated clothing before re-use.

In case of Inhaled contact:

Move person to fresh air and keep at rest until recovered. If any signs or symptoms occur or persist, get medical attention.

#### 5. FIRE FIGHTING MEASURES

This product is non-flammable, non-explosive and is combustible. Extinguish warehouse and factory fires using fine-water spray or foam. During a fire this product may release smoke and hazardous decomposition products. When fighting a major fire wear an air-supplied respirator. Do not allow fire-water to enter drains.

#### 6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

Procedure for spill:

- (1) Keep all bystanders away.
- (2) Wear full length clothing and butyl rubber gloves.



- (3) Re-position any leaking containers so as to minimize further leakage.
- (4) Dam and absorb spill with an absorbent material (e.g. sand or soil).
- (5) Shovel the absorbed spill into drums.
- (6) Disposal of the absorbed material will depend upon the extent of the spill.
  - For quantities up to 50L of product bury in a secure landfill site.
  - For quantities greater than 50L seek advice from the manufacturer (use emergency contact number below) before attempting disposal. Contain in a secure location until disposal method is established.
- (7) Decontaminate the spill area with detergent and water and rinse with the smallest volume of water practicable

## 7. HANDLING AND STORAGE

Stable under normal conditions of use. Store bottle in carton to protect from light and store below 30°C. Store in tightly sealed original containers in a dry secure place away from fertilisers, seed, feed and food. Store out of direct sunlight. Keep out of reach of children, unauthorised persons and animals.

## 8. EXPOSURE CONTROL/PERSONAL PROTECTION

### Exposure standards

Ingredient Occupational Exposure Limit

Abamectin 40 µg/m<sup>3</sup> (Internal exposure control limit during manufacture/formulation of the product. A NOHSC exposure standard has not been assigned to abamectin.)

Mineral Oil Mists (Paraffinic Oil): 5 mg/m<sup>3</sup> (NOHSC – time weighted average)

Butylated Hydroxytoluene: 10mg/m<sup>3</sup> (NOHSC – time weighted average)

No official exposure standards have been set for the other ingredients of this product.

### Engineering controls

No special requirements when product is used outdoors.

When opening the container, preparing spray and using prepared spray. Wear:

- Cotton overalls buttoned to the neck and wrist
- A washable hat
- Elbow-length butyl rubber gloves
- Goggles

For space applications, wear:

- Cotton overall buttoned to the neck and wrist
- A washable hat
- Elbow-length butyl rubber gloves



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- Googles
- Half facepiece respirator with dust cartridge or canister

## 9. PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                        |                                |
|------------------------|--------------------------------|
| Appearance:            | Colorless to yellow liquid     |
| Flash point:           | 71°C (closed cup)              |
| Odor:                  | Hexanol odor                   |
| Flammability:          | Non flammable                  |
| Boiling/Melting point: | 150-155°C (AI)                 |
| Combustibility:        | Combustible C1                 |
| Vapor pressure:        | <2 x 10 <sup>-4</sup> mPa (AI) |
| Solubility in water:   | 7 – 10µg/L at 20°C (AI)        |
| Specific gravity:      | 0.90 to 0.92 at 25°C           |

## 10. STABILITY AND REACTIVITY

Storage stability between 5°C and 50°C, state permissible, storage time under given storage condition indicating the range of temperature. Qualified. Stable at room temperature.

## 11. TOXICOLOGICAL INFORMATION

### Information on toxicological effects

#### **ACUTE TOXICITY**

##### Swallowed: **HARMFUL**

Tests indicate this product is harmful following single doses of a similar formulation.

(LD50 = approx. 325 mg/kg)

##### Dermal absorption: **NOT HARMFUL**

Tests on rabbits indicate this product is not harmful following skin contact with a similar formulation (LD50 > 2000 mg/kg)

##### Inhaled: **HARMFUL**

Tests indicate this product is harmful due to inhalation of a similar formulation.

(LC50 (4 h) = 3.5mg/L)



**Note:** Pure abamectin is considered highly toxic in acute animal studies. The acute oral toxicity studies of abamectin have shown clear differences among species in sensitivity to abamectin toxicity. Rodents are uniquely sensitive compared to the other species in which the compound has been tested. It is therefore inappropriate to base human risk assessment on the response in rodents.

Eye Irritation: **SLIGHT IRRITANT**

Skin Irritation: **SLIGHT IRRITANT**

Skin Sensitization: **NOT A SENSITISER**

## 12. ECOLOGICAL INFORMATION

### Toxicity

Adopt good working practices, so that the product is not released into the environment.

Data referred to the mixture, abamectin:

#### **Fish – Acute toxicity:**

Highly toxic to fish and aquatic invertebrates

Rainbow trout,

LC50 (96 h): 3.2 ppb

Bluegill sunfish,

LC50 (96 h): 9.6 ppb

Sheepshead minnow,

LC50 (96 h): 15 ppb

Channel catfish,

LC50 (96 h): 24 ppb

Crap,

LC50 (96 h): 42 ppb

Daphnia magna (a small freshwater crustacean),

LC50 (48 h): 0.34 ppb

Pink shrimp (*Panaeus duorarum*),

LC50 (96 h): 1.6 ppb

Mysid shrimp,

LC50 (96 h): 0.022 ppb

Eastern oysters,

LC50 (96 h): 430 ppb

Blue crab,

LC50 (96 h): 153 ppb



Abamectin did not bio-accumulate in bluegill sunfish exposed to 0.099 ppb for 28 days in a flow-through tank. On day 28, the concentration of residues in the fish was 6.8 ppb, but this rapidly decreased to 0.32 ppb by day 42. The BCF value calculated from this study is 52, indicating that abamectin does not accumulate or persist in fish.

**Birds – Acute toxicity:**

Bobwhite quail,

LD50: 2,000 mg/kg

LC50 (5 days): 3,102 ppm

Mallard ducks,

LC50 (5 days): 383 ppm

There were no adverse effects on reproduction when mallard ducks were fed dietary doses of 3, 6 or 12 ppm for 18 weeks.

**Earthworm – Acute toxicity:**

LC50 (28 days): 28 ppm

Earthworms will not be adversely affected by use of abamectin at recommended application rates.

**Honey bee – Acute toxicity:**

Highly toxic to bees

LC50 (24 h): 0.002 ug/bee

LD50: 0.009 ug/bee

Rapid degradation of abamectin will reduce the risk of bee deaths. Citrus and alfalfa foliage was not toxic to bees 24 to 48 hours after treatment with abamectin.

### 13. DISPOSAL CONSIDERATIONS

Dispose of empty, used containers by:

(a) Triple rinsing or preferably pressure rinsing containers with water. Add the rinsings to the spray tank.

DO NOT dispose of undiluted chemicals on site.

(b) If recycling, replace cap and return clean containers to recycler or designated collection point.

(c) If not recycling, break, crush or puncture and bury empty containers in a local authority landfill. If no landfill is available, bury the containers below 500 mm in a disposal pit specifically marked and set up for this purpose clear of waterways, desirable vegetation and tree roots. Empty containers and product should not be burnt.



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#### 14. TRANSPORT INFORMATION

UN number: 2902

Class: 6.1

Packing Group: III

Dangerous goods class: 6.1

Poisons schedule: 6

#### 15. REGULATORY INFORMATION

Risk Phrases:

R20: Harmful by inhalation.

R38: Irritating to skin.

Safety Phrases:

S2: Keep out of reach of children

S13: Keep away from food, drink and animal feeding stuff

S25: Avoid contact with eyes

S36: Wear suitable protective clothing

#### 16. OTHER INFORMATION

The above information is believed to be correct based on our present knowledge but does not purport to be complete. For research use only by trained personnel. The burden of safe use of this material rests entirely with the user.